

Rant or Rave: Variation over Time in the Language of Online Reviews

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1. Introduction

Online reviews have become a vital resource for consumer decision-making. As e-commerce and social media rapidly grow, consumers increasingly depend on others' experiences to reduce purchase risks and fill information gaps. Surveys show that over 90% of consumers read online reviews before making a purchase, with most dedicating at least a minute to them. This demonstrates the growing impact of reviews on shaping buying choices. For platforms and businesses, reviews not only influence sales and reputation but also play a crucial role in recommendation algorithms and market competitiveness.

However, the language used in reviews is not fixed. Over time, the way people express themselves in online reviews has changed significantly. For example, Ziser, Webber & Cohen (2023), in their study “Rant or Rave: Variation over Time in the Language of Online Reviews,” found that on platforms like Yelp, IMDb, and Amazon, reviews have become shorter, more polarized, and increasingly depend on frequent sentiment words.

Studying these changes is essential for several reasons. From the consumer point of view, review length, sentiment, and lexical diversity impact perceptions of informativeness and trustworthiness. Shorter or overly biased reviews may lessen their usefulness or even mislead readers. From the perspective of businesses and platforms, review quality directly influences user experience and platform credibility; understanding how review language has evolved can help develop more effective recommendation and ranking systems. Academically, online review data offers valuable insights into language development and human–computer interaction.

Prompted by these reasons, this study aims to systematically analyze the linguistic features of reviews over time, focusing on Amazon’s Electronics category from 2008 to 2023. Our goal is to partly replicate and expand on the work of Ziser, Webber & Cohen (2023), using more recent data to examine ongoing trends. Specifically, we seek to answer these research questions:

- RQ1: How has the way sentiment is expressed in online reviews changed from 2008 to 2023?
- RQ2: What are the long-term trends in review length and polarization?
- RQ3: Has the use of extreme sentiment words become more concentrated?

- RQ4: How do positive and negative reviews differ in these patterns of change?

2. Data retrieval

The data used in this study comes from the Amazon Product Reviews 2023 dataset, which was compiled and publicly released by the McAuley team at the University of California, San Diego (UCSD). This dataset spans multiple categories from 1996 to 2023 and includes over 500 million reviews, making it very large. We focused on the Electronics category and looked at four specific years: 2008, 2013, 2018, and 2023.

We chose the four time points of 2008, 2013, 2018, and 2023 because, compared to yearly analysis, these key years help us identify research trends and impacts while lowering computational costs and complexity. The period from 2008 to 2023 covers 15 years, providing enough time to observe long-term changes in comment languages. Selecting data points roughly every five years allows us to capture the phased nature of linguistic shifts.

Due to the large volume of raw data, processing all reviews directly was unfeasible because of storage and computational constraints. Therefore, we employed random sampling to reduce the dataset. Specifically, we used the streaming mode from the Hugging Face datasets library to read reviews sequentially, filtered data by comment timestamps to target specific years, and within each year, randomly sampled 10,000 reviews using reservoir sampling to ensure representativeness. During data cleaning, we removed corrupted and empty rows and standardized timestamp formats. We ultimately retained the following core fields:

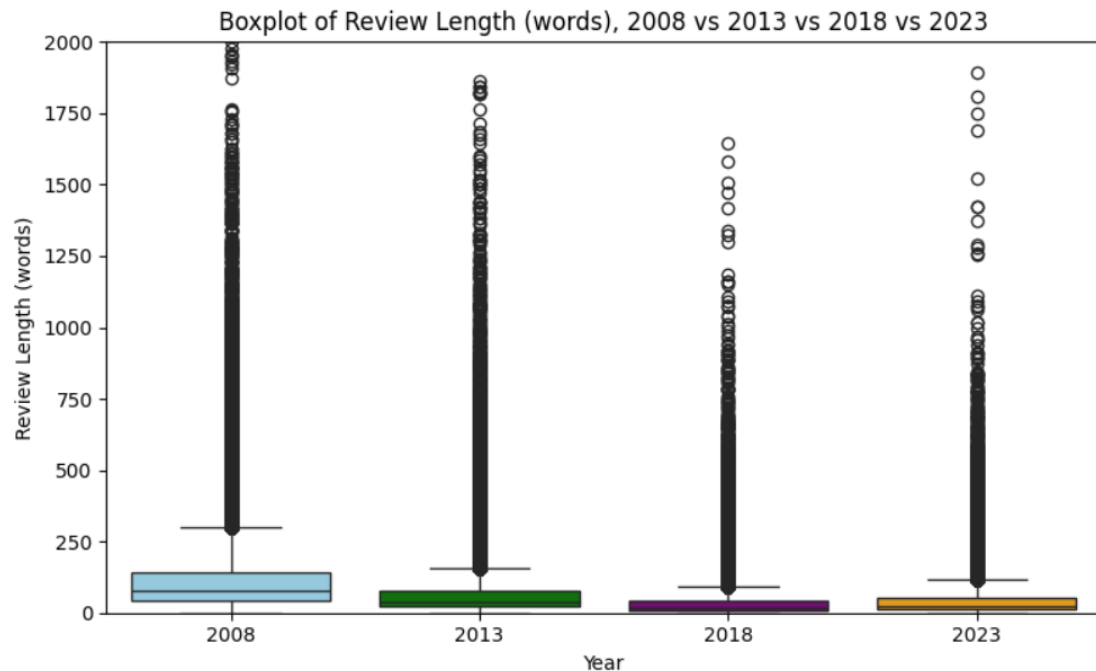
- ♦ rating (star rating, 1–5 stars)
- ♦ title (review title)
- ♦ text (review body)
- ♦ user_id (user identifier)
- ♦ asin (product identifier)
- ♦ timestamp (comment posting time)

This resulted in approximately 40,000 reviews (4 years $\times$ 10,000 reviews), serving as the foundation for further analysis.

3. Data Exploratory Analysis

The star rating distribution shows noticeable changes over time. In 2008 and 2013, more than 50% of reviews were 5-star, with a peak at 58.4% in 2013. In 2018, 65% of reviews rated 5 stars, while 1-star reviews accounted for 12%. By 2023, 5-star

reviews slightly declined to 62.3%, but 1-star reviews increased to 15.3%, indicating rising polarization. Overall, results indicate increasing polarization: more 5-star and 1-star reviews over time, while mid-range (2–4 stars) ratings decrease.



As shown in the graph, the length of comments has steadily decreased over the past 15 years. The median comment length was nearly 100 words in 2008, but by 2018 and 2023, it had dropped to about 40–50 words, with a more concentrated distribution. Overall, comment language has gradually become more extreme and concise over time.

4. Methods

4.1 Sentiment Analysis (VADER)

I used the VADER (Valence Aware Dictionary for Sentiment Reasoning) lexicon-based method to analyze each review individually. For each review, I calculated and stored sentiment scores, including the compound (overall polarity), pos (proportion of positive words), and neg (proportion of negative words). This provided sentiment scores for all reviews over four years, which I used to identify overall trends. Later, I grouped the reviews based on star ratings (1–2 stars as Negative, 4–5 stars as Positive) to compare positive and negative reviews.

4.2 Language diversity and comprehensiveness

To analyze the use of extreme sentiment words, we extracted high-intensity terms

from the VADER lexicon (valence ≥ 3 or ≤ -3) and tokenized review texts for word-level matching. We computed the following metrics: (1) the proportion of extreme sentiment words among all tokens; (2) their proportion among sentiment-bearing tokens; (3) the diversity of sentiment word types and the coverage of the top 1% most frequent sentiment words. These statistics were aggregated for each year (2008, 2013, 2018, 2023) and further broken down by Positive and Negative reviews. Additionally, we visualized the overall (ungrouped) share of extreme words across years and grouped (positive/negative) extreme word share across years.

4.3 Review Length and One-Sidedness Analysis

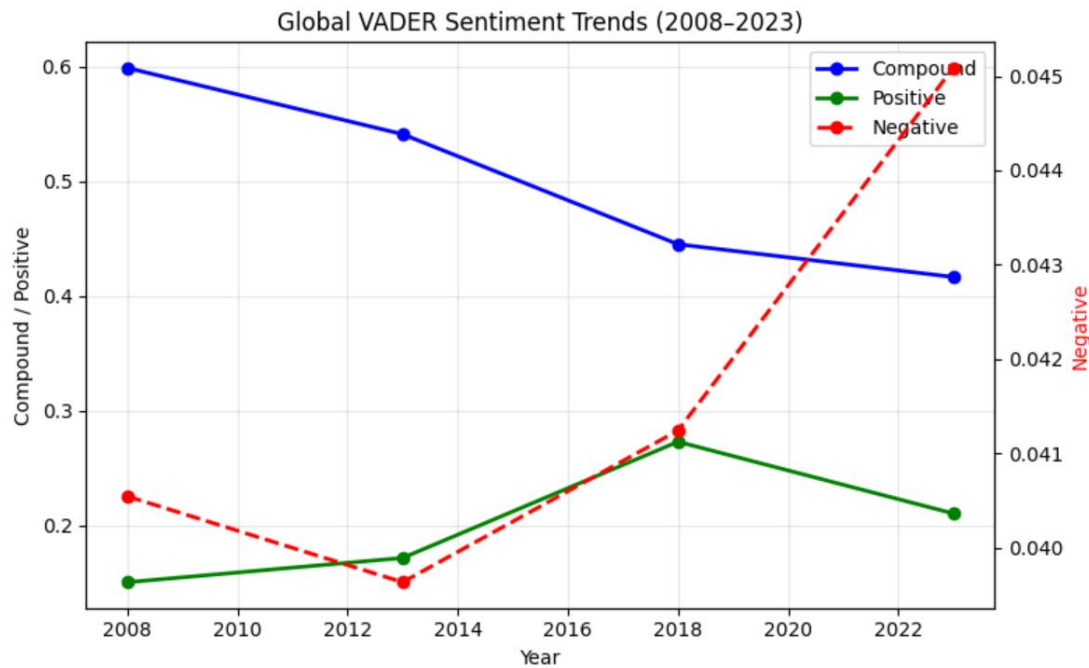
We analyzed review length and one-sidedness using a lexicon-based approach. Using the VADER lexicon, we built sets of positive and negative words, including common negation markers (e.g., not, never, don't). When a negation appeared within a short window before a sentiment word, we reversed its polarity. Each review was tokenized to count positive and negative words, helping us identify one-sided reviews—either positive reviews with only positive words or negative reviews with only negative words. Review length was measured in words, and averages were calculated for each year and sentiment category (Positive/Negative). Finally, we plotted yearly trends of average length, one-sidedness, and the concentration of extreme sentiment words to observe changes in review language from 2008 to 2023.

4.4 Illustrative Examples of Extreme Sentiment Words

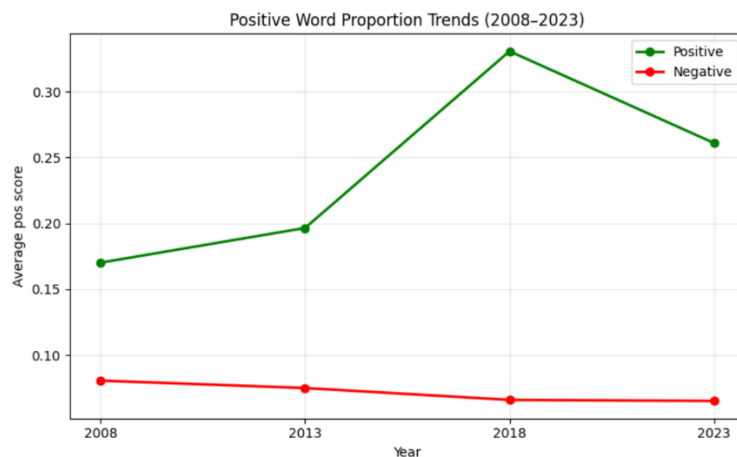
To supplement the overall statistics, we selected sample reviews containing extreme sentiment words. We then picked the shortest reviews with at least one such word, limiting the sample to five examples for each class (Positive vs. Negative) and year (2008, 2023). This demonstrates how even very brief reviews can still express strong emotional polarity through the use of extreme sentiment words.

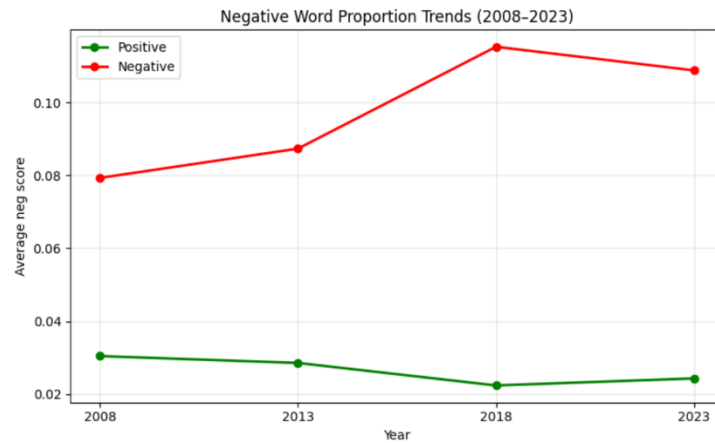
5. Results

5.1 Sentiment Analysis (VADER)



The figure above illustrates how global VADER sentiment scores for Amazon Electronics reviews have shifted from 2008 to 2023. The compound score clearly declines, dropping from about 0.60 in 2008 to around 0.42 in 2023, indicating that overall sentiment has become less positive over time. The share of positive words increased steadily until 2018 (peaking at roughly 0.27), but then declined again in 2023 (to about 0.21). Conversely, the proportion of negative words (red line) slightly decreased from 2008 to 2013 but then rose sharply, reaching its highest level in 2023 (above 0.045).



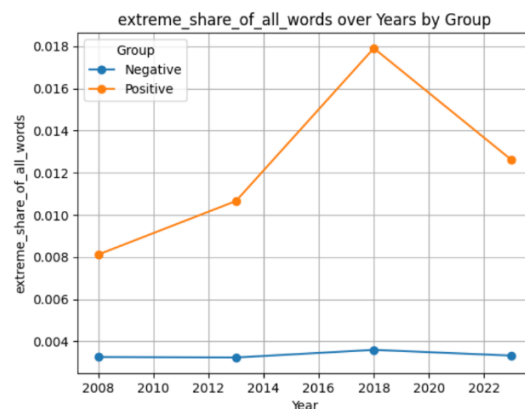
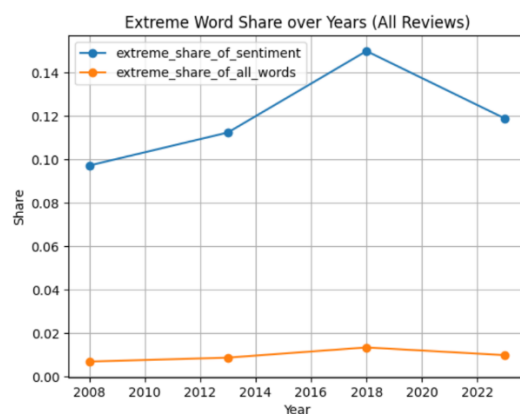


The above figure shows that the average compound scores of reviews dropped over time: positive reviews became less positive, while negative reviews became more negative. Figure 2 shows that the percentage of positive words increased until 2018, then decreased sharply. On the other hand, Figure 3 shows that the percentage of negative words in negative reviews steadily increased, peaking around 2018 and remaining high after that.

Overall, from 2008 to 2023, the sentiment of Amazon Electronics reviews has gradually shifted toward negativity. The consistent decline in compound scores indicates that the overall tone of reviews is less positive than in earlier years. Notably, around 2018 marked a significant turning point: before this year, the share of positive words steadily increased while negative expressions remained relatively low; however, after 2018, positive sentiment dropped sharply, and negative sentiment rose quickly.

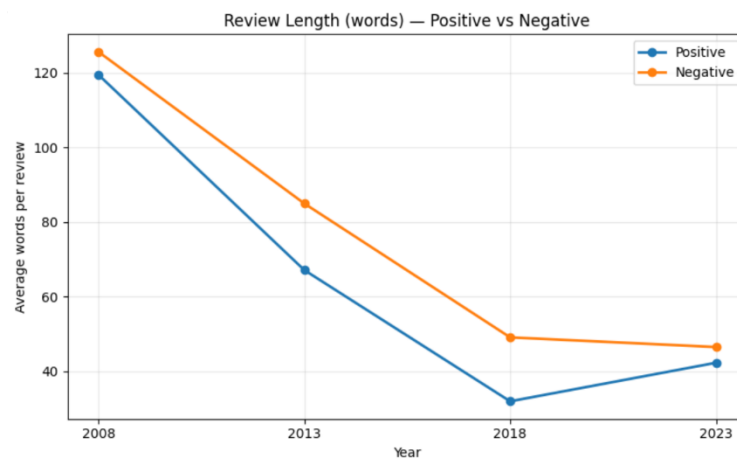
This suggests that consumers have become generally less satisfied with electronic products, which may be explained by rising standards and stricter expectations as the electronics industry has expanded.

5.2 Language diversity and comprehensiveness

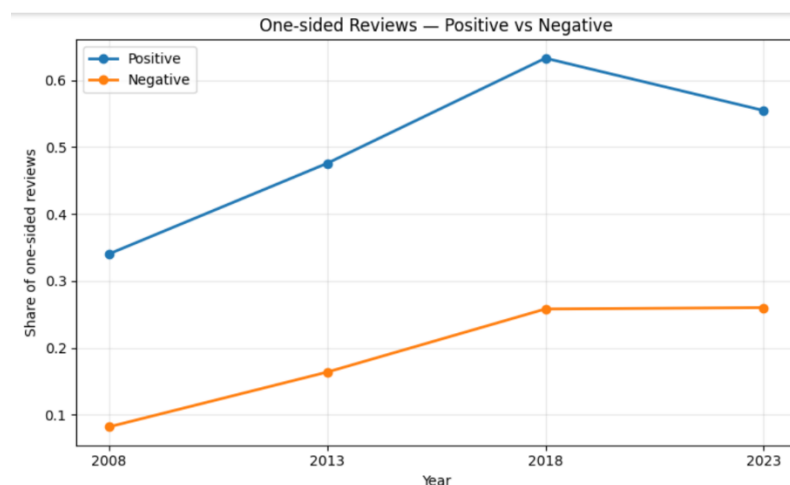


Our analysis of extreme sentiment word usage shows that over time, reviews have become more focused on a small set of very common extreme words, especially in positive reviews (e.g., great, love, best). This suggests a decrease in language variety: instead of using a wide range of descriptive words, reviewers are increasingly relying on a few repetitive, emotionally charged terms.

5.3 Review Length and One-Sidedness Analysis



Between 2008 and 2023, both positive and negative reviews became much shorter. Positive reviews shrank from about 120 words in 2008 to around 40 words in 2023, while negative reviews decreased from roughly 127 to 46 words. By 2023, the lengths of positive and negative reviews had become similar, showing a clear trend of less review detail over time.



The results in Figure X indicate that reviews have become more one-sided over time. For positive reviews, the proportion of one-sided cases steadily increased from about 34% in 2008 to a peak of 64% in 2018, then slightly declined to 55% in 2023. Negative reviews, although consistently less one-sided than positive ones, also grew significantly from around 8% in 2008 to 26% in 2018, and then

remained stable. This trend suggests that reviews are becoming less balanced, with users more likely to focus exclusively on either positive or negative aspects. Consequently, the depth and informativeness of reviews have lessened, as they tend to reflect a single emotional perspective rather than a detailed evaluation.

5.4 Illustrative Examples of Extreme Sentiment Words

=== 2008 Positive shortest intense reviews ===

	word_len	intense_terms	text
0	1	great	great
1	1	great	great
2	1	great	great
3	2	awesome	Awesome price!
4	2	great	Great product :)
5	2	great	Works great!
6	2	great	works great
7	4	great	GREAT PRODUCT FAST DELIVERY
8	4	great	Great RAM high speed.
9	4	great	Great product! Great service!

=== 2008 Negative shortest intense reviews ===

	word_len	intense_terms	text
0	13	dead	Dead on arrival. No answer at Mustek tech sup...

	word_len	intense_terms	text
1	13	worst	Sucks! Suxs! Linksys has the worst software ...
2	15	dead	Bought this mouse about fews months ago and it...
3	16	dead	Dead in 30 min. Sent back for full refund. A...
4	16	dead	Product only worked once, used it for 30min an...
5	16	worst	The worst pair of earphone I have ever owned. ...
6	17	worst	These are the worst earbuds I've ever experien...
7	17	worst	This router needs rebooted constantly. I have...
8	18	dead	This does not work reliably. It often shows t...
9	19	dead	Do not buy, not worth the effort to replace on...

=== 2023 Positive shortest intense reviews ===

	word_len	intense_terms	text
0	1	awesome	AWESOME
1	1	awesome	Awesome
2	1	awesome	Awesome
3	1	awesome	Awesome
4	1	awesome	Awesome

	word_len	intense_terms	text
5	1	awesome	Awesome
6	1	awesome	Awesome
7	1	awesome	Awesome
8	1	awesome	Awesome
9	1	awesome	Awesome

=== 2023 Negative shortest intense reviews ===

	word_len	intense_terms	text
0	1	worst	Worst
1	2	worst	Worst product
2	2	worst	Worst webcam
3	3	worst	Worst head phones
4	4	worst	The worst tv antenna.
5	4	worst	Worst flood light ever
6	4	dead	battery dead .after 2 month
7	5	killed	Killed my PSU! DONT BUY!!
8	5	worst	Worst reciever I've ever bought.

	word_len	intense_terms	text
9	6	dead	Bought 09/2022 dead 01/2023 Fan dead - System...

Results indicate that the use of extreme emotional words has become more simplified and focused over the past fifteen years. In 2008 reviews, while words like “great” and ‘worst’ were still common, they were usually part of full sentences with some explanation, such as “Dead in 30 min. Sent back for refund.” Conversely, 2023 reviews are very brief: positive comments are mostly just the word “Awesome,” and negative ones often simply say ‘Worst’ or “Dead,” with no detail or context. This suggests review language is becoming more concise and monotonous, with extreme terms used more prominently. Reviewers increasingly prefer expressing their attitudes directly through emotional words rather than providing detailed narratives.

6. Conclusion

Between 2008 and 2023, Amazon Electronics reviews experienced a clear transformation: reviews became notably shorter, more one-sided, and more uniform in language, with extreme sentiment words like great, awesome, worst, and dead dominating the discussion. Meanwhile, the overall sentiment shifted from mostly positive to a more negative outlook.

Several factors may explain these trends. The widespread use of mobile devices and social media has promoted shorter and more direct reviews. Increasing competition in the electronics market, along with rising consumer expectations, has made negative experiences more noticeable. Platform algorithms may also boost extreme and emotional content, encouraging users to adopt concise but stronger expressions. Finally, larger external shocks like the COVID-19 pandemic may have increased dissatisfaction and sped up the shift toward more negative reviews.

7. Critique and reflection

This study has several limitations. First, we sampled only four years (2008, 2013, 2018, and 2023) instead of analyzing the entire time series. Although this approach provides a manageable scope and still captures long-term trends, it might miss fluctuations or turning points in intermediate years. Second, our analysis focused solely on the Amazon Electronics category, which could limit the applicability of the findings to other domains or platforms. Future research could expand the temporal

scope, include multiple product domains, and utilize more advanced models (e.g., deep learning–based sentiment analysis) to provide a more detailed understanding of how online review language evolves.

Literature

Ziser, Y., Webber, B., & Cohen, S. (2023). *Rant or rave: Variation over time in the language of online reviews*. Transactions of the Association for Computational Linguistics, 11, 1–17.